#manydaughters - Pre-Analysis Plan

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1. Sample and Inclusion Criteria

In this study, we will test if the support for gender equality (measured by attitudes or outcomes) and the political preferences of individuals are affected by whether they have daughters.

Therefore, we use data from the Socio-Economic Panel (SOEP). The years considered for each hypothesis test depend on data availability and are described below in the hypotheses. The SOEP is a nationally representative longitudinal household panel survey that is collected through interviews. Currently, around 30,000 respondents from around 22,000 households are surveyed each year.

Regardless of the specific inclusion criteria defined below, we include all persons who have at least one child (the variable of the SOEP biol/lb0285: number of children needs to be different from 0 at least one time in considered years). Additionally, we only include individuals where there is information about the gender of all children defined as "son/daughter ". Furthermore, we only consider subjects for whom we have data on the respective variables that we include for each hypothesis test.

There are about 30,000 observations per year, according to the information on the SOEP homepage. If 29% of the included individuals have at least one child defined as their "Son/daughter" there should be about 8700 observations per year.[1]

The data from the survey questions that we use to test the hypotheses are identical (at least in terms of content) in all surveys of the SOEP. The phrasing is based on source questionnaire files posted on the SOEP webpage. We describe each dependent variable, the main independent variables, and all the control variables included in each hypothesis separately.

1 According to the Mikrozensus 2023 there are 29% of households that have at least one child living in the household in 2013 ((https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Bevoelkerung/Haushalte-Familien/Publikationen/Downloads-Haushalte/statistischer-bericht-mikrozensus-haushalte-familien-2010300237005-erstergebnisse.html?utm_source=chatgpt.com downloaded on 06-30-2025).

2. Strategy for Hypothesis Testing

We will test the following hypotheses in the study, and we will interpret a two-sided p-value <0.05 in the below hypotheses tests as "suggestive evidence" and a two-sided p-value <0.01 "as statistically significant evidence". Whenever we have such an effect for the independent variable, we accept the corresponding hypothesis. Unless otherwise stated all hypotheses will be tested in an OLS regression in which the variables described below are used. In all four analyzes we will cluster standard errors on the "Never Changing Person ID" ("pid") variable for analysis on the individual level (H1, H2, H4) and on the "Current Wave Household Number" (hid) for analysis on the household level (H3). In the following, we describe our hypotheses and indicate the variables considered in the model.

3. Hypothesis

3.1 Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes).

3.1.1 Regression Model and Dependent Variable

Regression Model: OLS with cluster standard errors at the individual level ("pid")

Dependent variable: We include the following questions to measure support for gender equality based on attitudes. To build an index that is used as our dependent variable in the regression analysis, we first standardize the variables, adding the standardized values of each of the six variables and dividing by 6. The resulting variable is again z-standardized. We assume that these variables are highly correlated and, consequently, are suited for building an index variable.

The following questions are assessed on a 7-point scale from 1 (do not agree at all) to 7 (totally agree)

1. Having a job is the best way for a woman to be independent. (pl/plj0621)
2. Even a married woman should have a paid job so that she can be financially independent. (pl/plj0622)
3. If a woman earns more money than her partner, this inevitably leads to problems. (pl/plj0623)
4. For parents, vocational training or higher education for their sons should be more important

than vocational training or higher education for their daughters. (pl/plj0624)
5. At home, the husband should have the final say. (pl/plj0625)

The following question is assessed on a 0-10 Likert scale with 0 (should definitely not happen in a democracy) to 10 (should definitely happen in a democracy).

6. Women have the same rights as men. (pl/plj0620)

We recode questions 3, 4, and 5 in the other direction so that for all variables higher values indicate higher support for gender.

As these questions are only included in years 2016 to 2018, and at least one of our control variables is only assessed in 2016, we consider 2016 for our analysis. In the following, we refer to this index as the "**attitude toward gender equality**".

As an additional inclusion criterion for Hypothesis 1, respondents must report at least one child in 2016, as indicated by a value greater than zero on the SOEP variable biol/lb0285 (Number of Children).

3.1.2 Independent Variable

Our main explanatory variable is "having a daughter", coded as a binary indicator (1 = has a daughter; 0 = does not have a daughter), constructed using the SOEP variables biobirth/kidpnr and biobirth/kidsex.

To evaluate Hypothesis 1, a positive coefficient on this variable is interpreted as evidence that having a daughter is associated with stronger support for gender equality. Statistical significance is assessed using standard errors clustered at the individual level.

3.1.3 Control Variables

In the analysis we control for three types of variables: (i) personal characteristics of the individual; (ii) characteristics of the household the individual is living in; (iii) preferences and attitudes of the individual. In the following, we describe these variables in more detail.

Personal Characteristics

As part of the survey section used to collect information about all household members, information about “gender” (variable code in the SOEP dataset "ppathl/sex; we recode gender so that "0" = male and "1" = female) and “year born” (pl/ple0010_h) are collected for all individuals. We use these to calculate respondents' age and apply a log transformation of age in the analysis, as we expect a non-linear (downward) association with age. We also include a binary variable indicating whether the respondent was born in Germany (pl/plj0010, 1 = yes, 2 = no).

For “education”, we use the SOEP variable pgen/pgpsbil to represent the respondent’s school-leaving qualification, which we recode into a three-category variable: “no degree” (codes 6, 7, 8); “compulsory school with degree” (codes 1, 2); “continuing school or high school” (codes 3, 4).

We also control for parental education “education_father” and “education_mother”, based on the father's (biol/lb0380) and mother's (biol/lb0381) school-leaving qualifications. Both are recoded using the same three-category scale as described above.

Household Characteristics

In the analysis, we also control for “number of children in the household” (biobirth/sumkids). We build a dummy variable “singleparent”, that is constructed using the variables “partner in household” (pl/pld0133, 1 = yes, 0 = no). We additionally include the variable “daughter’s distance” (pl/plj0122_h), which takes on the value 1 if the daughter lives in the same household or the same house. We also want to include the personal contribution to household income and therefore use the ratio of “current monthly net labor income (Euro)” (pgen/pglabnet) as numerator to “monthly household net income (Euro)” (hl/hlc0005_h) as denominator. This will be our variable “income-ratio”. We expect that this measure might interact with gender and thus introduce a second variable that is our interaction term “income-ratio*gender”.

Preferences and Attitudes

We include a variable indicating whether the respondent is religious, based on the SOEP item “Belonging to a church or religious community” (pl/plh0258_h). This variable is coded as 1 if the respondent reports affiliation with a religious community, and as 0 if they state no religious affiliation or provide no response.

3.2 Hypothesis 2: Having daughters leads to more left-leaning political preferences.

3.2.1 Regression Model and Dependent Variable

We apply a difference-in-differences (DiD) approach with individual and year fixed effects to examine the effect of having a daughter on political attitudes. The DiD is based on repeated observations before and after the treatment (birth of a daughter). Inclusion in the treatment group requires that a respondent has a daughter born between 2006 and 2018, and that we observe their political preferences both before and after the birth. We define the *pre-treatment period* as the last wave before the birth of the daughter, and the *post-treatment period* as the first wave after the birth of the daughter. Therefore, the years considered for a household may differ.

Subjects without post-treatment observations are retained in the dataset but do not contribute to the estimation of the treatment effect, as is standard in panel DiD regressions with individual fixed effects.

The validity of causal inference depends in the DiD on the parallel trend assumption**.** To assess the plausibility of this assumption, we explore pre-treatment trends in the outcome variable across groups. Depending on data availability and sample size, we may implement event-study specifications using years relative to the timing of the treatment event (birth of the first daughter). These exploratory analyses are intended to inform the interpretation of DiD estimates but will not affect the primary specification of the regression model.

Dependent Variable: we use the question about **general political attitude** (pl/plh0004) on a 11 point Likert-scale from left (0) to right (10) as our main dependent variable:

"In politics, people sometimes talk of "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?"

This variable is assessed in years 2005, 2009, 2014, and 2019.

3.2.2 Independent Variable

Birth year of the first daughter. We construct this variable using the SOEP variables biobirth/kidpnr (child number), biobirth/kidegeb (year of birth), and biobirth/kidsex (sex of the child).

To evaluate the hypothesis, we estimate the interaction effect between having a daughter and the post-treatment period. We expect a negative coefficient, indicating that the birth of a daughter is associated with a shift toward more left-leaning political preferences. Statistical significance is assessed using standard errors clustered at the individual level.

3.2.3 Control Variables

As control variables we consider the same variables as in Hypothesis 1: individual characteristics ("gender", "ln(age)", "born in Germany", "level of education", "education_mother", "education_father"), household characteristics ("number of children", "single parent", "income ratio", and "income-ratio × gender"), and individual preferences and attitudes ("religious") and also account for political interests, using the SOEP variable *pl/plh0007*, which is originally measured on a 4-point Likert scale where higher values indicate lower interest in politics. For ease of interpretation, we reverse-code this variable so that higher values reflect greater political interest**.**

3.3 Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)

3.3.1 Regression Model and Dependent Variable

To measure support for gender equality based on behaviors or outcomes, we construct the "**Household Equality Gap**". This measure captures the deviation from an equal division of unpaid work between men and women in a household.

To do this, we examine the unpaid work ratio as follows:

We sum the hours spent by each parent within one household on housework (pli0043_h/pli0012_h/pli0016_h), childcare (pli0044_h/pli0019_h/pli0022_h), errands (pli0054), and repairs/gardening (pli0049_h).

The **Unpaid Work Ratio** is defined as:

UnpaidWorkRatio = Total hours (woman) / Total hours (man)

If the man reports zero hours of unpaid work, we assign the UnpaidWorkRatio a value of 0.01. This avoids undefined values due to division by zero. Then, the ratio is built by dividing the hours spent by women by the hours spent by men.

We then calculate the Equality Gap as the absolute deviation from perfect equality:

Equality Gap = | 1 – UnpaidWorkRatio |

Lower values of the Equality Gap indicate a more balanced division of unpaid work between partners.

We estimate this outcome using ordinary least squares (OLS) regression, including year fixed effects to control for time trends. Standard errors are clustered at the household level (hid).

The analysis includes all survey years between 2000 and 2020 in which all relevant variables are available. We restrict the sample to households in which a man and a woman live together as parents, identified using the SOEP variable pld0133 ("partner in household", 1 = yes). As an additional inclusion criterion, we include only households where the number of children living in the household is greater than 0. To calculate this variable, we use the distance to son/daughter (pl/plj0122_h and pl/plj0123_h, codes 0 and 1).

3.3.2 Independent Variable

The key independent variable is identical to that used in Hypothesis 1: a binary indicator for whether the household has a daughter, based on the SOEP variables biobirth/kidpnr and biobirth/kidsex.

To evaluate the hypothesis, we interpret a negative coefficient on the daughter dummy as evidence that having a daughter is associated with a smaller Equality Gap, i.e., a more gender-equal division of unpaid household labor. Statistical significance is assessed using standard errors clustered at the household level.

3.3.3 Control Variables

We consider the average age of the parents in the household (biobirth/gebjahr) and the difference between the ages of both parents. In addition, we will control for household income (hl/hlc0005_h) and the number of children in the household (biobirth/sumkids). If the data is available for both parents, we also want to check whether both parents are religious (pl/plh0258_h).

We additionally control for hours spent on occupational work (pli0038_h) and also introduce an interaction term with gender.

3.4 Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

3.4.1 Regression Model and Dependent Variable

We estimate an ordinary least squares (OLS) regression with standard errors clustered at the individual level (pid). As in Hypothesis 1, the dependent variable is the respondent's attitude toward gender equality, measured in the year 2016.

The inclusion criteria are identical to those defined in Hypothesis 1.

3.4.2 Independent Variable

Our explanatory variables are "having a daughter" (as defined in Hypothesis 1), gender, and the interaction between having a daughter and gender term (daughter × gender). The coefficient on the interaction term (daughter × gender) captures the additional effect of having a daughter for women relative to men. A positive coefficient indicates that the effect of having a daughter is stronger for women than for men. Conversely, a negative coefficient implies that the effect is stronger for men, which is consistent with the hypothesis that men increase their support for gender equality more than women when having a daughter.

3.4.3 Control Variables

We control for the same variables as in H1: (i) personal characteristics of the individual, except for gender as it is part of the main independent variables; (ii) characteristics of the household the individual is living in, (iii) preferences and attitudes of the individual.